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RESEARCH ARTICLE

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Microstimulation of V1 affects the detection of visual targets: manipulation of target contrast

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Abstract Electrical microstimulation of the striate cortex (area V1) in monkeys delays the execution of saccadic eye movements generated to a visual target located in the receptive field of the stimulated neurons. We have argued that this effect is because of disruption of the visual signal transmitted along the geniculostriate pathway. The delivery of electrical stimulation to V1 evokes a punctate light or dark phosphene in human subjects. If electrical stimulation of V1 in monkeys evokes a light or dark phosphene, then one might expect that the delay effect might vary according to whether monkeys are required to detect a light or a dark visual target. For instance, if the stimulation is activating V1 elements coding for a light visual stimulus but not a dark visual stimulus then stimulation may delay saccades generated to a light target but not to a dark target. We tested this idea by having monkeys generate saccadic eye movements to light or dark visual targets immediately after the stimulation was delivered to V1. We found that the delay effect induced by stimulation varied with target contrast, but remained invariant to whether a bright or dark visual target was presented in the receptive field of the stimulated neurons. The significance of these results is discussed with regard to using monkeys to develop a visual prosthesis for the blind.

Keywords Occipital cortex · Luminance contrast · Electrical microstimulation · Phosphenes · Visual prosthesis · Rhesus monkeys

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Introduction

Electrical microstimulation of striate cortex (area V1) of monkeys interferes with the selection and detection of visual targets when the stimulation is delivered concurrently with the presentation of the visual targets (Schiller and Tehovnik 2001; Tehovnik et al. 2002; Tehovnik and Slocum 2003a; Slocum and Tehovnik 2004). Also stimulation of V1 before the presentation of a visual target delays the subsequent execution of a visually guided saccade to the target (Tehovnik et al. 2004). This delay effect occurs when the visual target is situated in and confined to the receptive field of the stimulated neurons. In fact, the delay field is precisely in register with the receptive field. The best sites yielding the delay effect are located between 1 mm and 2 mm below the cortical surface when using currents under 100 μA. We have proposed that the delay effect is due to disruption of the visual signal transmitted along the geniculostriate pathway (Tehovnik et al. 2005).

We have determined the excitability properties of the directly stimulated elements in monkey V1 mediating the delay effect (Tehovnik et al. 2004). The elements exhibited chronaxies (which are an estimate of the excitability of the directly stimulated fibers) ranging between 0.1 ms and 0.3 ms.[1] These values overlap with the range of chronaxies reported for cortical pyramidal fibers (Stoney et al. 1968; Asanuma et al. 1976). Simple and complex cells in V1 have a pyramidal morphology (Gilbert and Wiesel 1979) and these cells are readily activated by electrical microstimulation (Ronner et al. 1981). The chronaxie values of V1 elements mediating the delay effect in monkeys overlap with those mediating phosphenes in humans (Brindley and Lewin 1968a; Dobelle and Mladejovsky 1974). Also excitation of V1 in humans is known to mask the perception of visual

[1] A chronaxie is the shortest duration of an effective electrical stimulation pulse having strength equal to twice the minimum strength required for neuronal excitation at the longest pulse durations.

stimuli (Kammer et al. 2005). This suggests that whenever stimulation produces a delay effect a phosphene may be generated that masks the perception of visual stimuli in monkeys.

The size and visual-field location of a phosphene induced by electrical stimulation delivered to human V1 corresponds to the receptive-field size and retinotopic location of the stimulated neurons (Brindley and Lewin 1968a; Dobelle and Mladejovsky 1974; Hubel and Wiesel 1977; Schmidt et al. 1996). Phosphenes elicited by stimulation of human V1 can have diverse featural characteristics. Phosphenes, however, are typically described as white or black spots of light, circular in shape (Brindley and Lewin 1968a, b; Brindley 1972; Dobelle and Mladejovsky 1974; Lee et al. 2000).

Neurons in V1 are modulated by changes in luminance contrast, such that the higher the contrast the greater the responsivity of cells until the activity saturates at the highest contrasts (Tolhurst et al. 1981; Albrecht and Hamilton 1982; Tolhurst 1989; Sclar et al. 1990). These cells have contrast thresholds that can range from 5 to 80% contrast (Albrecht and Hamilton 1982; Sclar et al. 1990). Cells in V1, especially simple cells, respond best either to light increment or light decrement (Schiller et al. 1976). Hubel and Livingston (1990) have reported that cells responding to the same sign of contrast are clustered such that cells responding to light stimuli are segregated from those responding to dark stimuli.

We know from previous work (Tehovnik et al. 2004) that the size of a visual target situated in the receptive field of the stimulated neurons affects the magnitude of the delay effect; that is, once the target is larger than the receptive field of the stimulated neurons the stimulation no longer induces a delay. However, it has not yet been determined whether features other than the size of the visual stimulus affect this delay. Experiments were therefore conducted to see if the delay effect evoked at a given V1 site varies according to the contrast of the target positioned in the receptive field of the stimulated neurons.

Materials and methods

Subjects

Two adult rhesus monkeys (*Macaca mulatta*) C and M were used. Throughout this study food was freely available. A monkey's access to water was limited before each day of experimental testing. After testing, the monkeys were allowed to drink until sated before being returned to the vivarium. Monkeys were provided for in accordance with the National Institutes of Health Guide for the Care and Use of Laboratory Animals and the Guidelines of the Massachusetts Institute of Technology Committee on Animal Care.

Surgery

Monkeys were anesthetized intravenously with pentobarbital (20 mg kg^{-1}) and prepared for aseptic surgery. A scleral search coil was implanted (Judge et al. 1980), and a stainless-steel post, used to restrain the head, was secured to the skull with titanium screws and acrylic cement. Subsequently, a recording chamber was implanted over the right V1.

Behavioral tasks

A monkey, with head fixed in position, faced a computer monitor (Sony Multiscan E210) positioned 57 cm away. Monkeys performed all tasks in a dark room. Two types of behavioral task were performed – receptive-field mapping and a target-detection task. All trials started with the animal acquiring a fixation spot (0.1 degrees in diameter). During the subsequent fixation period, the monkey was required to keep his eyes within a 0.5 by 0.5 degree window; otherwise the trial was aborted.

The first task allowed us to map the visual receptive field of V1 cells at an electrode site (Fig. 1A, right). The monkey was required to remain fixated on the fixation spot as a bar of light was swept across the visual field. After termination of the bar stimulus, a target appeared at one of several locations and the monkey was required to generate a saccade to that location. The monkey was rewarded with a drop of apple juice for successful completion of this task. The location, orientation, speed, and size of the bar were varied while the receptive-field location was being determined. This location is typically specified as eccentricity and meridian, measured in degrees, relative to the center of gaze. The receptive-field location was subsequently used in the target detection task as the visual-field position at which one visual target was presented.

The second task is the target-detection task. A monkey was required to look at the fixation spot for 600 ms; otherwise the trial was aborted (Fig. 1B). Immediately after termination of the fixation spot, a light or dark circular visual target (0.2 degrees in diameter) was presented in the receptive-field location of the V1 cells under study or presented at one alternative location in the opposite quadrant. The monkey was given a juice reward for generating a saccade to the target within 500 ms after target onset (Fig. 1B, a). A 100-ms train of electrical stimulation was delivered to V1 on 50% of trials at the end of the fixation period (Fig. 1B, b). The stimulation commenced 100 ms before the end of the fixation period. Stimulation trials were used to determine the magnitude of the interference effect as evidenced by a decrease in the percent-correct score for saccades generated to the target in the receptive-field location and by an increase in the latency to evoke a saccade to this location. The two trial types, non-stimulation (control) and stimulation trials, were presented in random order.

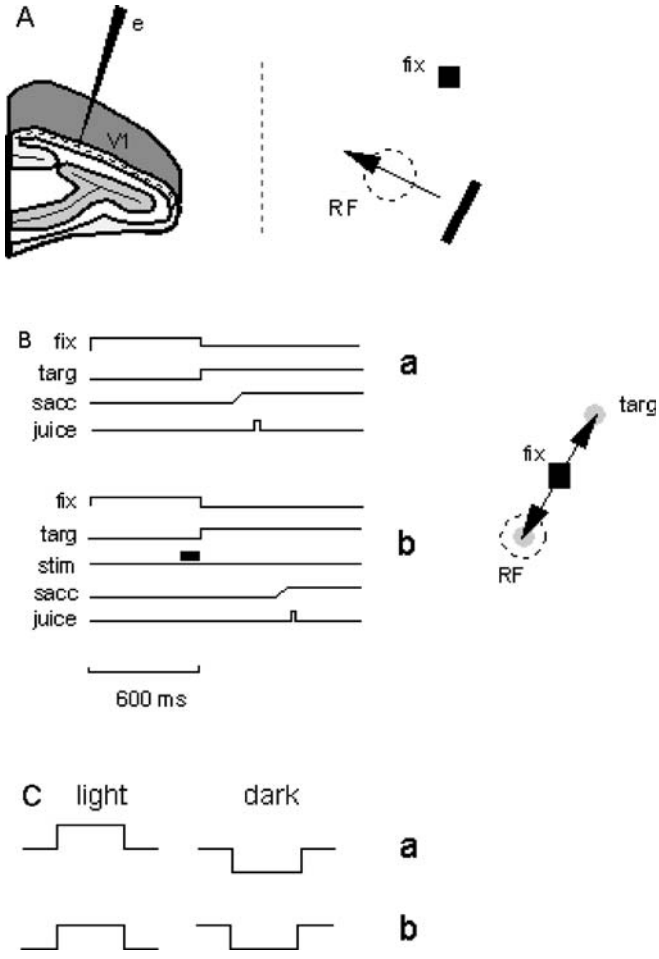


Fig. 1 Methods. **A**, *left* All electrodes (e) were lowered into V1 such that they penetrated V1 perpendicular to the cortical surface. **A**, *right* Receptive fields (RF) were mapped by moving a bar of light at different orientations across the visual field as the monkey fixated a spot (fix). **B** For all trials, an animal fixated a spot for 600 ms (*fix*) otherwise the trial was aborted and no juice reward delivered. Two types of trials were interleaved to study stimulation-evoked interference. (*a*) After fixation, a target (*targ*) was presented in the receptive-field location of the stimulated neurons or outside the receptive-field location after the extinction of the fixation spot (also see illustration to *right*). The animal was required to generate a saccade (*sacc*) to the target within 500 ms to obtain a juice reward (*juice*). (*b*) After fixation, a target (*targ*) was presented in the receptive-field location of the stimulated neurons or outside the receptive-field location after the extinction of the fixation spot. Before the end of the fixation period, a train of electrical stimulation (*stim*) was delivered to the neurons. The animal was required to generate a saccade (*sacc*) to the target within 500 ms to obtain a juice reward (*juice*). **C** The contrast polarity (i.e. light or dark) of a visual target was manipulated by either holding the background luminance level constant (*a*) or by interchanging the background luminance level with the target luminance level (*b*)

The contrast level of targets was measured using the Michaelson contrast equation: Percent contrast = [{target luminance − background luminance}/{target luminance + background luminance}]×100%. The background luminance of the Sony monitor was typically set at 76.7 cd m^{-2}. All our experiments were conducted with the background luminance within the photopic range of vision, always greater than or equal to 76.7 cd m^{-2}. Two methods were used to vary contrast polarity: the background luminance of the monitor was fixed while the light and dark contrast levels of the targets were varied (Fig. 1C, a) or the background luminance of the monitor was varied while holding the contrast level of the light and dark targets constant (Fig. 1C, b).

Estimating electrode depth

For all electrode penetrations made into cortex, a standard method was used to deduce the electrode depth with respect to the top of V1 (Tehovnik et al. 2002; Tehovnik and Slocum 2003b). At the beginning of each test session, an electrode was lowered into V1 (Fig. 1A, left). When single units were encountered on a pass, the electrode was withdrawn by 2 mm and left to settle. After waiting 30 min, the electrode was advanced again slowly while a bar of light was swept across the visual field of the monkey. Using an audio-monitor and an oscilloscope, unit responses were detected immediately upon entering cortex. After mapping the receptive field, the electrode was further adjusted in depth until the unit activity was just detectable. Such adjustments were continued intermittently over a 2 h period to guarantee that the electrode was well settled at the top of V1. After 2 h, this point marked the initial depth at the top of V1. At the end of a test session, the depth at the top of V1 was remeasured by noting the point at which units were just barely responsive to visual stimuli as the electrode was slowly withdrawn from the brain. This yielded the final depth at the top of V1. The estimated depth at the top of V1 was taken as the average of the initial and final depth estimates.

Data collection and analysis

A PDP-11/73 computer controlled the presentation of visual stimuli, the delivery of electrical stimulation, the collection and storage of multiple unit activity, the collection and display of eye position (sampled at 200 Hz), and the delivery of juice reward.

Electrical microstimulation and unit recording

Constant-current charge-balanced biphasic pulses were delivered to the brain via a monopolar glass-coated platinum-iridium electrode (at 0.5–1 MΩ tested at 1 kHz) using a Grass S88 stimulator attached to a pair of constant-current stimulus isolation units (Grass PSIU6B, Quincy, MA, USA). For each biphasic pulse an anodal and a cathodal pulse followed in immediate succession to produce anode-first pulses. Both pulses of a pair had the same amplitude and duration. Current was measured by voltage drop across a 1,000 Ω resistor

placed in series with the return lead of the stimulator. The current was monitored continuously using an oscilloscope and was read as the amplitude of one pulse (anodal or cathodal) of a biphasic pair. The stimulation conditions used while monkeys performed the target detection task were: the pulse duration, train duration, and frequency were fixed at 0.2 ms, 100 ms, and 200 Hz, respectively. For each experiment the current was fixed and could range from 15 μA to 100 μA.

Electrodes were introduced perpendicularly to the cortical surface with a hydraulic microdrive (Fig. 1A, left). The action potentials were amplified (Bak A-1B, Germantown, MD, USA) and filtered (Krohn-Hite 3750, Horsham, PA, USA), isolated with a window discriminator (Bak DIS-1, Germantown, MD, USA), and displayed on an oscilloscope (Tektronix TDS-210, Beaverton, OR, USA). The window settings of the discriminator were adjusted to detect multiple units. The computer was used to count the number of multiple units occurring over a 100-ms period after the onset of a visual target situated in the receptive field of the V1 cells under study. On trials in which no target appeared the units were sampled over the same 100-ms period.

Results

Stimulation sites

All electrode penetrations into V1 were normal to the cortical surface (Fig. 1A, left). For monkey C the penetrations contained cells with receptive-field centers varying from 3.2 degrees to 4.1 degrees of eccentricity and from 217 degrees to 227 degrees of meridian. For monkey M, the penetrations contained cells with receptive-field centers varying from 3.9 degrees to 5.5 degrees of eccentricity and from 223 degrees to 249 degrees of meridian. A total of 10 and 20 penetration sites were studied in monkeys C and M, respectively. For each penetration, one depth from 0.9 mm to 2.0 mm below the cortical surface was chosen for study of V1. However, in two cases deeper penetrations were made to depths of 3–4 mm yielding sites located in V2.

General effects

Electrical stimulation at the end of the fixation period was found to increase the latency of saccadic eye movements generated to a bright target centered in the receptive field of the stimulated neurons (data from one site: Fig. 2A, bottom). The increase in latency was observed for targets whose contrast varied from 10 to 30%. No increase in latency was observed when the target was situated outside the receptive-field location (Fig. 2B,

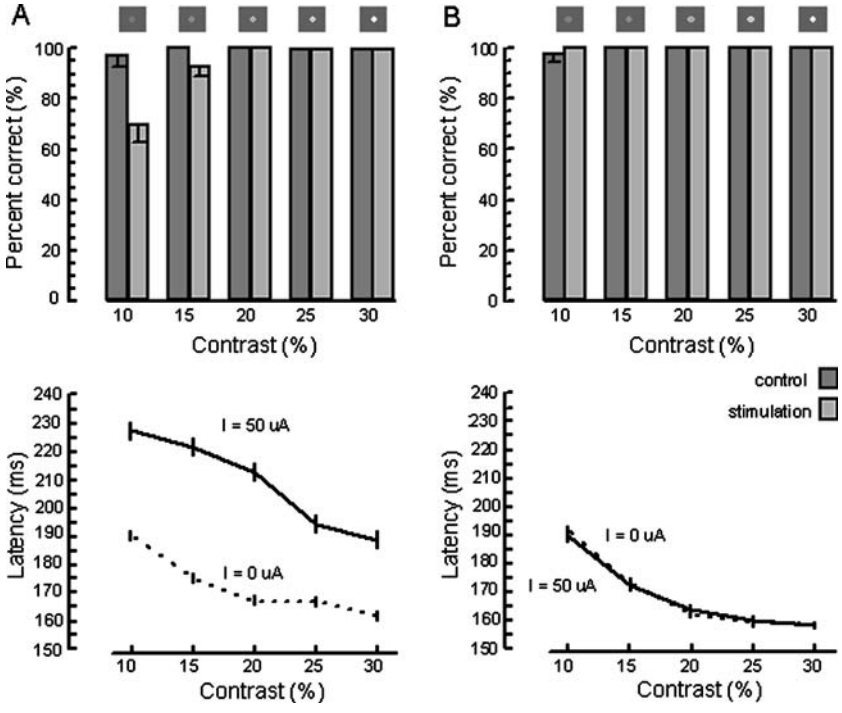


Fig. 2 Effect of electrical stimulation on detection of a target presented in the receptive field of the stimulated neurons (**A**) and on detection of a target presented outside the receptive-field location (**B**). **A**, **B**, *top* Percent correct is plotted as a function of target contrast for stimulation and control trials. Each value is based on 40 trials. Standard errors are shown. **A**, **B**, *bottom* Latency of saccades is plotted as a function of target contrast for stimulation and control trials. Each value is based on 40 trials. Standard errors are shown. Stimulation current was fixed at 50 μA. The depth of stimulation was at 1.3 mm below the cortical surface. The receptive-field location of the units at the stimulation site was at 242 degrees of meridian and at 4.2 degrees of eccentricity. The target used was brighter than background. For other details see Materials and methods and Fig. 1B. Data from monkey M

bottom). In addition, stimulation affected the percent-correct performance of the animal when low-contrast targets were used, but only when the target was situated in the receptive field of the stimulated neurons (Figs. 2A, B, top). In these cases, the stimulation often caused foreshortening of a saccade to the receptive field target such that the end-point of the saccade fell short of the target window (and hence no reward was delivered). The performance of an animal was always greater than chance (> 50% correct) and the animals typically generated a saccade toward the receptive-field target. This basic result was found for all sites studied as long as the stimulating electrode was positioned from 0.9 mm to 2.0 mm below the cortical surface.

On two occasions our electrode was located in the deepest layers of V2 (as evidence by an abrupt increase in receptive-field size and a change in receptive-field position as the electrode was pushed beyond and below V1). Electrical stimulation here never produced an increase in saccadic latency or a drop in the percent-correct score using currents as high as 100 μA.

Saccadic latency increases with current

Electrical stimulation had a differential effect on the latency to evoke saccades to the target located in the receptive field of the stimulated neurons depending on the current level used (data from one site: Fig. 3C). Electrical stimulation (using 100 μA) produced a drop in the percent-correct score (Fig. 3A) and an increase in latency (Fig. 3B) for saccades generated to a bright target situated in the receptive-field location of the stimulated neurons. The drop in percentage correct occurred when the target was of low contrast only and the increase in saccadic latency occurred for all target contrasts tested. A latency difference between stimulation and control trials for saccades generated to the receptive-field target was determined for different current levels. The greater the current level the greater the latency difference for all target contrasts tested.

Delay effects using different polarities of target contrast

We tested to see if the contrast polarity of the visual target in the receptive field of the stimulated neurons produces a differential effect on saccadic latency. For a bright target the contrast level at which the maximum difference in saccadic latency was observed between stimulation and control trials was determined. For the example shown (Fig. 4A), the contrast level at which a maximum latency difference was observed for a bright target occurred at 25% contrast (Fig. 4B, arrow). Subsequently, the monkey was required to generate a saccadic eye movement to a light or dark target whose contrast was 25% of background. The contrast polarity was set for the light and dark target by switching between the target and background luminance levels across

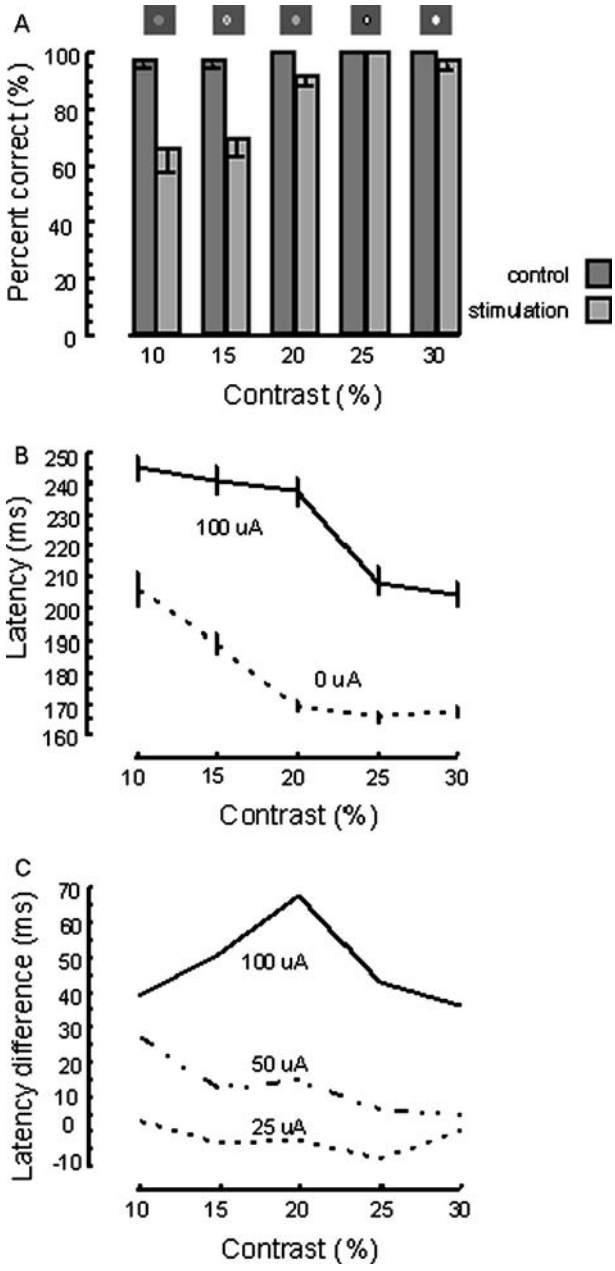


Fig. 3 Effect of electrical stimulation on detection of a target presented in the receptive field of the stimulated neurons. **A** Percentage correct is plotted as a function of target contrast for stimulation and control trials. Each value is based on 40 trials. Standard errors are shown. **B** Latency of saccades is plotted as a function of target contrast for stimulation and control trials. Each value is based on 40 trials. Standard errors are shown. **C** Latency difference between stimulation and control trials is plotted as a function of target contrast for three levels of current: 25, 50, and 100 μA. Data for the 100 μA condition are derived from (**B**). For all experiments the depth of stimulation was 1 mm below the cortical surface. The receptive-field location of the units at the stimulation site was at 239 degrees of meridian and at 4.3 degrees of eccentricity. The target used was brighter than background. For other details see Materials and methods and Fig. 1B. Data from monkey M

blocks of trials (see Fig. 1C, b). For half the blocks ($n = 5$) the light target was centered on a dark background whose luminance was set to that of the dark

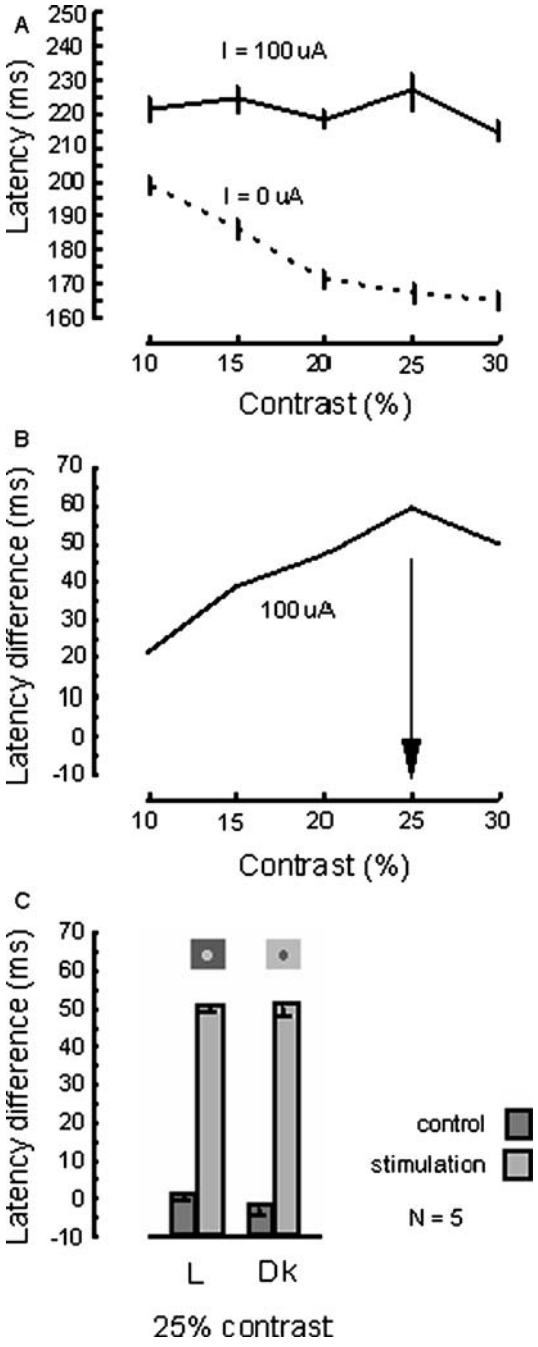


Fig. 4 Effect of electrical stimulation on detection of a target presented in the receptive field of the stimulated neurons. **A** Latency of saccades is plotted as a function of target contrast for stimulation and control trials. Each value is based on 20 trials. Standard errors are shown. **B** Latency difference between stimulation and control trials is plotted as a function of target contrast. The *arrow* indicates the contrast level for which there was the largest latency difference. Data derived from (**A**). **C** Latency difference is plotted as a function of whether the target was light (*L*) or dark (*Dk*) at 25% contrast for stimulation and control trials. To make contrast equivalent for the light and dark target conditions the background and target contrast levels were interchanged: for the light target condition the background was dark and the target was light at 25% contrast and for the dark target condition the background was light and the target was dark at 25% contrast (see Fig. 1C, b). Each value in the figure is based on five blocks of 20 trials per block. Standard errors are shown. For all experiments, current was fixed at 100 μA and the depth of stimulation was at 1.3 mm below the cortical surface. The receptive-field location of the units at the stimulation site was at 243 degrees of meridian and at 4.3 degrees of eccentricity. For other details see Materials and methods and Fig. 1B. Data from monkey

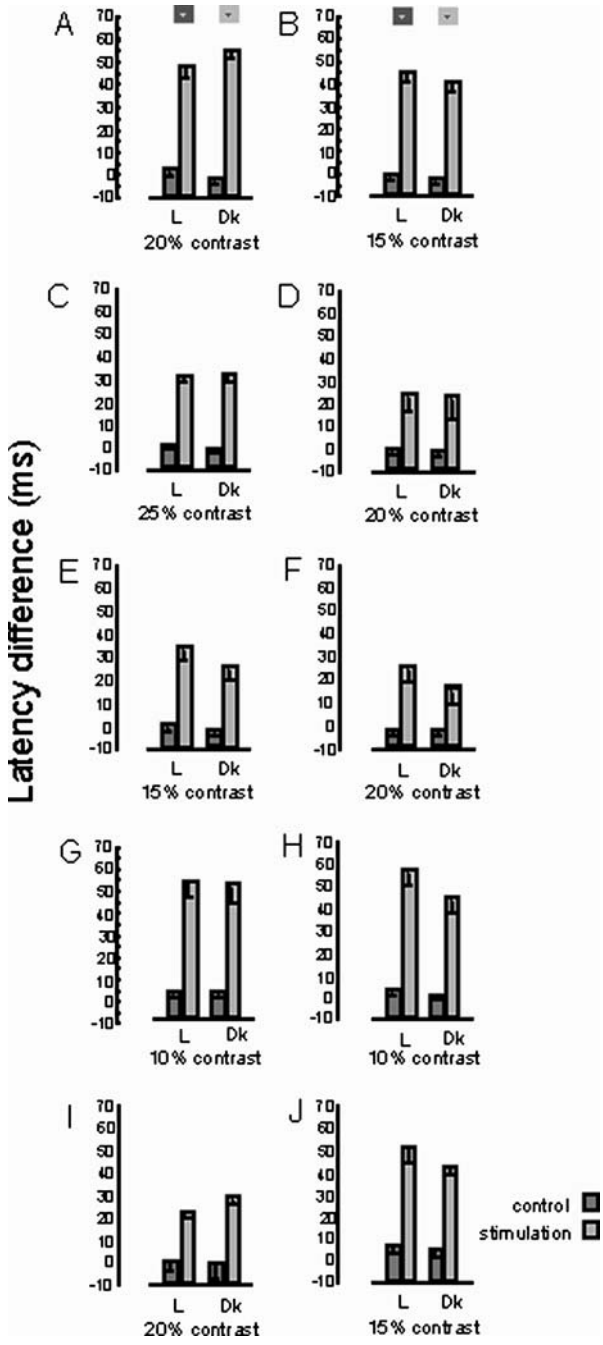
target; for the remaining blocks ($n = 5$) the dark target was centered on a light background whose luminance was set to that of the light target. Electrical stimulation did not differently affect the increase in latency for saccades generated to light and dark targets of comparable contrast (Fig. 4C, $t(4) = 0.3$, $P > 0.05$). This null result ($P > 0.05$) was found for all 12 additional sites tested (Figs. 5, 6), each located 1–2 mm below the cortical surface using currents of 15–100 μA. For two of these stimulation sites (Fig. 6), the responsivity of cells was assessed by presenting a light or dark target in the receptive field of the neurons under study. The net spike count of the cells (measured over a 100-ms period following target onset) remained invariant to the contrast polarity of the target.

Finally, a series of experiments was conducted in which the background luminance of the monitor was fixed while the contrast levels of the target varied from −30 to +30% contrast (Fig. 1C, a). The negative values represent dark targets and the positive values represent light targets. Electrical stimulation had a comparable effect on increases in saccadic latency for dark and light targets (Fig. 7).

Current spread estimates

Current delivered to neural tissue through a microelectrode spreads effectively proportional to the square root of the current divided by the square root of the current-distance constant (Tehovnik 1996). For large pyramidal tract cells, this constant can range from 300 μA mm^{-2} to 3000 μA mm^{-2} with an average of about 1,000 μA mm^{-2} (Stoney et al. 1968). These values were computed using a cathodal current pulse having a duration of 0.2 ms. The current-distance constant reflects the excitability of a neural element 1 mm away from the electrode tip: an element having a constant of 1,000 μA mm^{-2} would require a 1,000 μA current to be activated 1 mm away 50% of the time. The greater the current-distance constant the less the conduction velocity of an axonal element (Jankowska and Roberts 1972; Roberts and Smith 1973; Hentall et al. 1984). Therefore the size of a neuron's axon, and whether it is myelinated or not, are related to the current-distance constant.

To estimate the spread of current in V1, we will assume a current-distance constant of 1,000 μA mm^{-2} . This is a conservative estimate for V1 given that the neuronal elements in V1 of primates tend to be smaller and therefore less excitable than those of other parts of cerebral cortex (Cragg 1967; Rockel et al. 1980; O'Kusky and Colonnier 1982; Fries 1984; Peters 1987). Indeed



◀

Fig. 5 Effect of electrical stimulation on the detection of targets of opposite contrast polarity for stimulation delivered to ten different V1 sites (**A** through **J**) with **A** to **F** from monkey M and with **G** to **J** from monkey C. At each site the percentage contrast level tested was that with the greatest latency difference between stimulation and control trials. See Fig. 4 for details. For each site, latency difference is plotted as a function of whether the target was light (*L*) or dark (*Dk*) for stimulation and control trials. To make contrast equivalent for the light and dark target conditions the background and target contrast levels were interchanged: for the light target condition the background was dark and the target was light at given percentage contrast, and for the dark target condition the background was light and the target was dark for the same contrast condition (see Fig. 1C, b). Each value in a figure is based on 5 blocks of 20 trials per block. Standard errors are shown. The currents used were fixed for a site and ranged from 15 μA to 100 μA; the depths tested ranged from 1 mm to 2 mm below the cortical surface. For other details see Materials and methods and Fig. 1B

the conduction velocities of pyramidal neurons exiting V1 tend to be significantly lower than those of large pyramidal neurons (Finlay et al. 1976; Macpherson et al. 1982). Using the equation, $r=(I/K)^{1/2}$, where r is the distance of effective current spread from the electrode tip in mm, I is the current used in μA, and K is the current-distance constant in μA mm^{-2}, a 15 to 100 μA current (the range used in this study) delivered to V1 is estimated to activate elements within 122 to 316 μm, respectively, from the electrode tip. These values are comparable with those obtained using behavioral methods (Tehovnik et al. 2004).

Discussion

Consistent with our previous study (Tehovnik et al. 2004), microstimulation of V1 before execution of a saccadic eye movement increased the latency of a saccade generated to a punctate visual target (i.e. 0.2 degrees of visual angle) located in the receptive field of the stimulated neurons. When the visual target was situated outside the receptive field of the stimulated neurons there was no delay in the execution of visually guided saccades. The increase in saccadic latency due to V1 stimulation occurred over the full range of target contrasts tested, i.e. 10–30% contrast, with the maximum increases occurring at a particular contrast level. For the lowest contrast tested (i.e. 10%), stimulation caused degradation in performance: a monkey's saccades undershot the position of the target located in the receptive field of the stimulated neurons. We suspect that the electrical stimulation never "blinded" an animal, because the performance of an animal was always greater than chance and because an animal typically generated a saccade toward the receptive-field target despite the stimulation.

We conducted a series of tests wherein we manipulated the contrast polarity of the target falling in the receptive field of the stimulated neurons. Given that electrical stimulation of V1 evokes both black and white phosphenes in humans and that neurons in macaque V1 that respond preferentially to light increment or light decrement are segregated (Brindley and Lewin 1968a, b; Brindley 1972; Dobelle and Mladejovsky 1974; Schiller et al. 1976; Hubel and Livingston 1990; Lee et al. 2000), we wanted to see if electrical microstimulation in V1 differentially affects the detection of bright and dark targets. We manipulated the contrast polarity of targets in two ways—by switching the luminance of the target with the luminance of the background (Figs. 4, 5, 6) or by holding the luminance of the background constant and varying target luminance (Fig. 7). Both methods

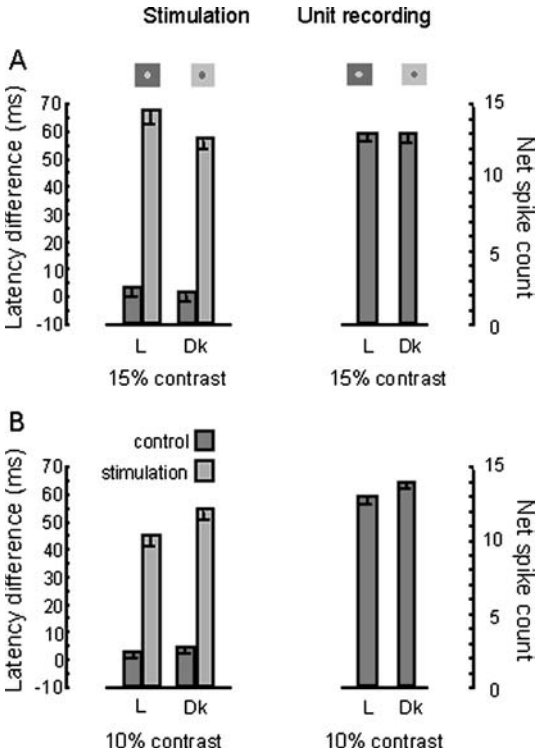


Fig. 6 The effect of electrical stimulation on the detection of targets with different contrast polarities (*Stimulation*) and the net discharge of cells at the stimulation sites to the targets having the different contrast polarities (*Unit recording*). (*Left*) Effect of stimulation on the detection of targets of opposite contrast polarity for stimulation delivered to two different V1 sites (**A** and **B**). At each site, the percentage contrast level tested was that with the greatest latency difference between stimulation and control trials. For each site, latency difference is plotted as a function of whether the target was light (*L*) or dark (*Dk*) for stimulation and control trials. The current level used was 20 and 25 μA in **A** and **B**, respectively. The depth of stimulation was 1.5 mm and 2.0 mm in **A** and **B**. See Fig. 4 for details. (*Right*) The responsivity of cells to the presentation of the light (*L*) or dark (*Dk*) targets is shown for neurons at stimulation sites **A** and **B**. Unit activity was collected for 100 ms after presentation of a visual target situated at the center of the receptive field (i.e. 217 degrees of meridian and 3.4 degrees of eccentricity for **A**, 227 degrees of meridian and 3.2 degrees of eccentricity for **B**). The net spike count during the 100-ms interval was determined by subtracting the spike count for blank trials from the spike count for target trials. Each bar graph is based on 20 trials. Standard errors are shown. Data from monkey C

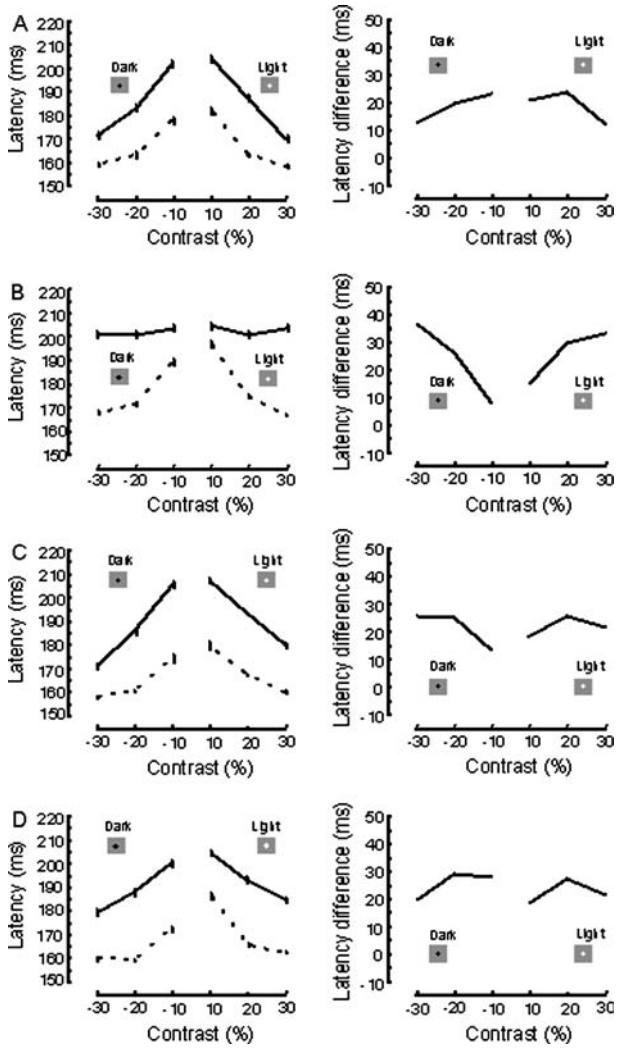


Fig. 7 Saccadic latency is plotted as a function of negative (*dark*) and positive (*light*) contrast levels for stimulation and control trials (*left*) and the corresponding latency difference between stimulation and control trials plotted as a function of contrast (*right*). Data from four V1 sites are shown (**A** through **D**). Within each left panel, the *solid curves* represent latency values from stimulation trials and the *dashed curves* represent latency values from control trials. Each latency value is based on 100 trials. Standard errors are shown. For each experiment, the current used and cortical depth tested were fixed. The currents ranged from 50 μA to 100 μA and the depths tested ranged from 0.95 mm to 1.65 mm below the cortical surface. The background luminance of the monitor was fixed for each experiment (see Fig. 1C, a). For other details see Materials and methods and Fig. 1B. All data from Monkey M

yielded similar results. The contrast polarity of the receptive-field target—i.e. whether the target was light or dark—had no effect on delaying visually guided saccades with microstimulation. Microstimulation induced a comparable increase in saccadic latency irrespective of whether the receptive-field target was light or dark. Thus electrical microstimulation of V1 over the range of currents used (i.e. 15–100 μA) indiscriminately activates neurons that mediate both the perception of light increment and light decrement.

This conclusion is also supported by our unit recording data showing that multiple units in V1 failed to discriminate between the presentation of light and dark targets. Our estimates (Tehovnik and Slocum 2003b) suggest that the region of V1 sampled with our low-impedance recording electrodes (0.5–1 MΩ tested at 1 KHz) was comparable to the region activated by our stimulation, i.e. within roughly 300 μm from the electrode tip.

Effect of V1 lesions compared to the disruptive effects of V1 microstimulation

Ablations of V1 abolish a monkey's ability to generate visually guided saccades to targets but spontaneously

generated saccades remain unaffected (Mohler and Wurtz 1977; Segraves et al. 1987; Zee et al. 1987). Many weeks after such an ablation, however, monkeys can detect and produce saccades to punctate visual targets even though saccadic accuracy and latency are slightly impaired. As with cortical ablations, V1 injections of ibotenic acid (which kills cells at the injection site) only temporarily abolish saccades generated to targets situated at the visual field represented by the killed cells (Newsome et al. 1985). When short-lasting agents such as lidocaine, muscimol, or bicuculine are injected into V1 monkeys fail to execute saccades to targets located in the affected part of the visual field almost immediately after the injection (Newsome et al. 1985; Schiller and Tehovnik 2003).

The disruptive effect induced by microstimulation of V1 differs from the effect of V1 lesions in two ways: (1) during stimulation monkeys can still generate saccadic eye movements in the direction of the affected field even though the saccades are foreshortened somewhat for the lowest contrast stimuli (e.g. 10% contrast), and (2) the monkeys never adapt to the disruptive effect of stimulation. On the last point, microstimulation of V1 induced an increase in saccadic latency no matter how often the stimulation was delivered to V1; this was true both within a daily session (which could include as many as 1,000 stimulations of the same site) and across sessions spanning many months of stimulation of sites representing the same quadrant of the visual field of a monkey.

The volume of tissue affected by V1 stimulation using currents between 25 μA and 100 μA was estimated to be from 0.064 to 0.418 mm^3, respectively, which represents 0.31 and 0.58 degrees of the visual field at an eccentricity of 4 degrees from the fovea (for details see Tehovnik et al. 2004). This volume is a fraction of that affected by injecting minute quantities of muscimol into V1 (0.9 mm^3 of tissue is inactivated by 0.5 μL muscimol at a concentration of 0.5 μg mL^{-1}; Schiller and Tehovnik 2003). Such injections of muscimol abolish all saccades made into the affected field of the drug-infused neurons. Perhaps higher currents (e.g. 160–360 μA) that affect a cortical volume comparable with that inactivated by muscimol might produce comparable deficits in the execution of visually guided saccades.

Phosphene induction and visual prosthetics

As already mentioned, when electrical stimulation is delivered to V1 in humans, white or black stationary spots of circular light are reported (Brindley and Lewin 1968a, b; Brindley 1972; Dobelle and Mladejovsky 1974; Schmidt et al. 1996; Lee et al. 2000). Our results indicate that stimulation of V1 using currents of 15–100 μA do not differentially affect saccades made to targets varying in contrast polarity even though V1 cells respond to light increment and light decrement (current study; Schiller et al. 1976; Hubel and Livingston 1990). The current levels used in the current investigation may have activated both "ON" and "OFF" elements thereby failing to differentiate between the two populations of cells. This conclusion is supported by our multiple-unit data (Fig. 6). To evoke phosphenes exhibiting visual features, currents below 10 μA might be necessary. Such low currents delivered through a microelectrode can evoke a colored phosphene (Schmidt et al. 1996).

An alternative explanation for our results is that the delay effect is not because of a disruption of the visual signal transmitted along the geniculostriate pathway (Tehovnik et al. 2005), but rather is because of a disturbance of the saccadic signal transmitted to the eye muscles. If the latter were true, selective stimulation of a right-eye or left-eye ocular dominance column in V1 as a visual target is presented separately to either eye should result in a comparable saccadic delay, because V1 efferents to the brainstem are not organized according to eye dominance (Finlay et al. 1976). When electrical stimulation was presented concurrently with the presentation of a visual stimulus it was found that the magnitude of interference was greatest when the visual target was presented to the eye with preferential innervation of the V1 column stimulated electrically (Slocum and Tehovnik 2004). We have yet to perform such a test on the delay effect observed in the current study.

When using microelectrodes, phosphenes elicited in human subjects are typically described as circular and stationary rather than as elongated and moving (Schmidt et al. 1996). The latter configuration might have been expected in humans given that single cells in V1 of monkeys are tuned to oriented and moving bars of light (Hubel and Wiesel 1977). The orientation and direction tuning of cells in V1 depends on γ-aminobutyric acid (GABA)ergic interneurons (Sillito 1975). For features such as edge orientation and edge motion to be realized with electrical stimulation, the stimulation might need to selectively activate these interneurons. It is possible that when electrical stimulation is delivered to V1, the stimulation preferentially activates the most excitable elements, the pyramidal neurons (Tehovnik et al. 2004), thereby bypassing the GABAergic circuitry. This might explain why phosphenes are often circular and stationary and not elongated and moving. If true, methods will need to be developed to activate GABAergic interneurons to elicit elongated and moving phosphenes.

Training monkeys to generate memory-guided saccades to visual field locations represented by V1 neurons stimulated electrically has now been demonstrated (Bradley et al. 2005). A next step will be to use the lowest current levels possible delivered through high-impedance electrodes to activate more selectively elements in V1 that encode visual features. This in combination with rigorous psychophysical testing of both monkeys and humans should accelerate the development of an effective visual prosthesis for the blind.

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